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STUDY MATERIAL

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RELATIONS

INTRODUCTION :

Let A and B be two sets. A binary relation or, simply, relation from A to B is a subset of A × B. Suppose R is a relation from A to B. Then R is a set of ordered pairs where each first element comes from A and each second element comes from B. That is, for each pair a ∈ A and b ∈ B, exactly one of the following is true :

(i) (a,b) ∈ R ; then we say "a is R-related to b", written aRb.

(ii) (a,b) ∉ R; then we say "a is not R-related to b", written a $\not R$ b

Ex. If A = {1, 2, 3} and B = {a, b, c}, then R = {(1, b), (2, c), (1, a), (3, a)} being a subset of A × B, is a relation from A to B. Here (1, b), (2, c), (1, a) and (3, a) ∈ R, so we write 1Rb, 2Rc, 1Ra and 3Ra. But (2, b) ∉ R, so we write 2 $\not R$ b

Void Relation : $\phi \subseteq A \times B$ and so it is a relation from A to B. This relation is called the void or empty relation from A to B.

Universal Relation : $A \times B \subseteq A \times B$ and so it is a relation from A to B. This relation is called the universal relation from A to B.

Total Number of Relations : Let A and B be two finite sets consisting of m and n elements respectively. Then A × B consists of mn ordered pairs. So, total number of subsets of A × B is 2^{mn}.

Domain and Range of a relation : Let R be a relation from a set A to a set B. Then the set of all first components or coordinates of the ordered pairs belonging to R is called to domain of R, while the set of all second components or coordinates of the ordered pairs in R is called the range of R.

Thus, Dom (R) = {a : (a, b) ∈ R}

and, Range (R) = {b : (a, b) ∈ R}

It is evident from the definition that the domain of a relation from A to B is a subset of A and its range is a subset of B.

Ex. Let A = {1, 3, 5, 7} and B = {2, 4, 6, 8} be two sets and let R be a relation from A to B defined by the phrase "(x, y) ∈ R ⇔ x > y". Under this relation R, we have

3R2, 5R2, 5R4, 7R2, 7R4 and 7R6

i.e. R = {(3, 2), (5, 2), (5, 4), (7, 2), (7, 4), (7, 6)}

∴ Dom (R) = {3, 5, 7} and Range (R) = {2, 4, 6}

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Inverse Relation : Let A, B be two sets and let R be a relation from a set A to a set B. Then the inverse of R, denoted by R^{-1}, is a relation from B to A and is defined by

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Clearly, $(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$

Also, $\text{Dom}(R) = \text{Range}(R^{-1})$ and $\text{Range}(R) = \text{Dom}(R^{-1})$

For example, let $A = \{1,2,3\}$ and $B = \{x,y,z\}$. Then the inverse of

$R = \{(1,y), (1,z), (3,y)\}$ is $R^{-1} = \{(y,1), (z,1), (y,3)\}$

Clearly, if R is any relation, then $(R^{-1})^{-1} = R$. Also, the domain and range of R^{-1} are equal, respectively, to the range and domain of R. Moreover, if R is a relation on A, then R^{-1} is also a relation on A.

Ex.1 Let A be the set of first ten natural numbers and let R be a relation on A defined by $(x, y) \in R \Leftrightarrow x + 2y = 10$, i.e. $R = \{(x, y) : x \in A, y \in A \text{ and } x + 2y = 10\}$. Express R and R^{-1} as sets of ordered pairs. Determine also (i) domain of R and R^{-1} (ii) range of R and R^{-1}

Sol. We have $(x, y) \in R \Leftrightarrow x + 2y = 10 \Leftrightarrow y = \dfrac{10-x}{2}$, $x, y \in A$

where $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Now, $x = 1 \Rightarrow y = \dfrac{10-1}{2} = \dfrac{9}{2} \notin A.$

This shows that 1 is not related to any element in A. Similarly we can observe. that 3, 5, 7, 9 and 10 are not related to any element of A under the defined relation

Further we find that :

For $x = 2$, $y = \dfrac{10-2}{2} = 4 \in A$ $\quad\therefore\ (2, 4) \in R$

For $x = 4$, $y = \dfrac{10-4}{2} = 3 \in A$ $\quad\therefore\ (4, 3) \in R$

For $x = 6$, $y = \dfrac{10-6}{2} = 2 \in A$ $\quad\therefore\ (6, 2) \in R$

For $x = 8$, $y = \dfrac{10-8}{2} = 1 \in A$ $\quad\therefore\ (8, 1) \in R$

Thus, $R = \{(2, 4), (4, 3), (6, 2), (8, 1)\}$

$\Rightarrow$ $R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$

Clearly, $\text{Dom}(R) = \{2, 4, 6, 8\} = \text{Range}(R^{-1})$

and, $\text{Range}(R) = \{4, 3, 2, 1\} = \text{Dom}(R^{-1})$

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Do yourself -1 :

1. Find the inverse relation of R = {(1,2), (3,4), (5,6)}.

2. Let R be a relation from A = {2,3,4,5} to B = {3,6,7,10} defined by R = {(a,b) | a divides b, a ∈ A, b ∈ B}, then number of elements in R^{-1} will be-

(A) 0 (B) 3 (C) 4 (D) 5

3. The R be a relation in the set N given by R = {(a, b) : a + b = 5, b > 1}. Which of the following will satisfy the given relation ?

(A) (2, 3) ∈ R (B) (4, 2) ∈ R (C) (2, 1) ∈ R (D) (4, 1) ∈ R

TYPES OF RELATIONS :

In this section we intend to define various types of relations on a given set A.

Identity Relation : Let A be a set. Then the relation $I_A = \{(a, a) : a \in A\}$ on A is called the identity relation on A.

In other words, a relation I_A on A is called the identity relation if every element of A is related to itself only.

Ex. The relation $I_A = \{(1, 1), (2, 2), (3, 3)\}$ is the identity relation on set A = {1, 2, 3}. But relations $R_1 = \{(1, 1), (2, 2)\}$ and $R_2 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$ are not identity relations on A, because $(3, 3) \notin R_1$ and in R_2 element 1 is related to elements 1 and 3.

Reflexive Relation : A relation R on a set A is said to be reflexive if every element of A is related to itself.

Thus, R on a set A is not reflexive if there exists an element $a \in A$ such that $(a , a) \notin R$.

Ex. Let A = {1, 2, 3} be a set. Then R = {(1, 1), (2, 2), (3, 3), (1, 3), (2, 1)} is a reflexive relation on A. But $R_1 = \{(1, 1), (3, 3), (2, 1), (3, 2)\}$ is not a reflexive relation on A, because $2 \in A$ but $(2, 2) \notin R_1$.

Note : Every Identity relation is a reflexive relation but every reflexive relation is not an identity relation.

Consider the following five relations on the set A = {1,2,3,4}

$R_1 = \{(1,1), (1,2), (2,3), (1,3), (4,4)\}$

$R_2 = \{(1,1) (1,2), (2,1), (2,2), (3,3), (4,4)\}$

$R_3 = \{(1,3), (2,1)\}$

$R_4 = \emptyset$, the empty relation

$R_5 = A \times A$, the universal relation.

Determine which of the relations are reflexive.

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Since A contains the four elements 1,2,3 and 4 a relation R on A is reflexive if it contains the four pairs (1,1), (2,2), (3,3) and (4,4). Thus only R_2 and the universal relation $R_5 = A \times A$ are reflexive. Note that R_1,R_3 and R_4 are not reflexive since, for example, (2,2) does not belong to any of them.

Symmetric Relation : A relation R on a set A is said to be a symmetric relation iff

$(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

i.e. $a R b \Rightarrow bRa$ for all $a, b, \in A$.

Note : Thus R is not symmetric if there exists $a,b \in A$ such that $(a,b) \in R$ but $(b,a) \notin R$

Ex. Let L be the set of all lines in a plane and let R be a relation defined on L by the rule $(x, y) \in R \Leftrightarrow$ x is perpendicular to y. Then R is a symmetric relation on L, because $L_1 \perp L_2 \Rightarrow L_2 \perp L_1$ i.e. $(L_1, L_2) \in R \Rightarrow (L_2, L_1) \in R$.

Ex. Let $A = \{1, 2, 3, 4\}$ and Let R_1 and R_2 be relation on A given by $R_1 = \{(1, 3), (1, 4), (3, 1), (2, 2), (4, 1)\}$ and $R_2 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$. Clearly, R_1 is a symmetric relation on A. However, R_2 is not so, because $(1, 3) \in R_2$ but $(3, 1) \notin R_2$

Transitive Relation : Let A be any set. A relation R on A is said to be a transitive relation iff

$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$

i.e. $a R b$ and $b R c \Rightarrow a R c$ for all $a, b, c \in A$

Note : Thus R is not transitive if there exist $a,b,c \in R$ such that $(a,b), (b,c) \in R$ but $(a,c) \notin R$.

Ex. On the set N of natural numbers, the relation R defined by $x R y \Rightarrow$ x is less than y is transitive, because for any $x, y, z \in N$

$x < y$ and $y < z \Rightarrow x < z \Rightarrow x R y$ and $y R z \Rightarrow x R z$

Ex. Let L be the set of all straight lines in a plane. Then the realtion 'is parallel to' on L is a transitive relation, because from any $\ell_1, \ell_2, \ell_3 \in L$.

$$\ell_1 \parallel \ell_2 \text{ and } \ell_2 \parallel \ell_3 \Rightarrow \ell_1 \parallel \ell_3$$

Antisymmetric Relation : Let A be any set. A relation R on set A is said to be an antisymmetric relation iff

$(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$

Note : Thus R is not antisymmetric if there exist distinct elements a and b in A such that aRb and bRa.

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Ex. Let R be a relation on the set N of natural numbers defined by

$$x\ R\ y \Leftrightarrow \text{'x divides y' for all } x, y \in N$$

This relation is an antisymmetric relation on N. Since for any two numbers $a, b \in N$

$$a|b \text{ and } b|a \Rightarrow a = b \quad \text{i.e. } a\ R\ b \text{ and } b\ R\ a \Rightarrow a = b$$

Asymmetric Relation : Asymmetric relation is the opposite of symmetric relation but not considered as equivalent to antisymmetric relation.

In set theory, A relation R on a set A is known as asymmetric relation if no $(b, a) \in R$ when $(a, b) \in R$ or we can even say that relation R on set A is asymmetric if only if $(a, b) \in R \Rightarrow (b, a) \notin R$.

An example of asymmetric relation is a "less than" relation between real numbers.

Equivalence Relation : A relation R on a set A is said to be an equivalence relation on A iff

(i) it is reflexive i.e. $(a, a) \in R$ for all $a \in A$

(ii) it is symmetric i.e. $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

(iii) it is transitive i.e. $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$.

Ex. Let R be a relation on the set of all lines in a plane defined by $(\ell_1, \ell_2) \in R \Leftrightarrow$ line ℓ_1 is parallel to line ℓ_2. R is an equivalence relation.

Note : It is not neccessary that every relation which is symmetric and transitive is also reflexive.

Equivalence class :

Let A be a set and threre is an equivalence relation R on A. then we define equivalence class of an element a in A, denoted by [a] as the set $\{x \in A : aRx\}$

Ex. $R = \{(1,1), (2,2), (1,2), (2,1)\}$

equivalence class of 1 and 2 are denoted by [1] and [2] respectively.

$[1] = \{1,2\}$

$[2] = \{1,2\}$

Note : $xRy \Leftrightarrow [x] = [y]$

Ex. Let $A = \{1,2,3,4,5\}$ and relation $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$. Define equivalence class of each element of A.

Sol. Given relation R is an equivalence relation

$[1] = \{1,2\}$

$[2] = \{1,2\}$

$[3] = \{3\}$

$[4] = \{4,5\}$

$[5] = \{4,5\}$

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Partial ordering relations :

A relation R on set A is said to be partial order relation iff

(1) it is a reflexive relation

(2) it is an anti-symmetric relation

(3) it is a transitive relation

Ex. Consider the following five relations on the set A = {1,2,3,4}

R_1 = {(1,1), (1,2), (2,3), (1,3), (4,4)}

R_2 = {(1,1) (1,2), (2,1), (2,2), (3,3), (4,4)}

R_3 = {(1,3), (2,1)}

$R_4 = \emptyset$, the empty relation

R_5 = A × A, the universal relation.

Determine which of the above relations are partial order relation

Sol. None of the given relation is a partial order relation.

Do yourself -2 :

1. Let L be the set of all lines in a plane and R be the relation in L defined as R = {(L_1,L_2) : L_1 is perpendicular to L_2}. Show that R is symmetric but neither reflexive nor transitive.

2. Show that the relation R in the set {1,2,3} given by R = {(1,1),(2,2), (3,3), (1,2), (2,3)} is reflexive but neither symmetric nor transitive.

3. Show that the relation R in the set Z of integers given by R = {(a,b) : 2 divides a – b} is an equivalence relation.

4. Let R be the relation defined in the set A = {1,2,3,4,5,6,7} by R = {(a,b) : both a and b are either odd or even}. Show that R is an equivalence relation. Further, show that all the elements of the subset {1,3,5,7} are related to each other and all the elements of the subset {2,4,6} are related to each other, but no element of the subset {1,3,5,7} is related to any element of the subset {2,4,6}.

5. Which of the following is not a partial order relation -

(A) R_1 = {(a, b) : a, b ∈ z, a < b}

(B) R_2 = {(a, b) : a, b ∈ z, a ≤ b}

(C) R_3 = {(a, b) : a, b ∈ z, a = b}

(D) None of the above

6. Let A = {1, 2, 3, 4, 5} and R = {(1, 2), (2, 1), (4, 5), (5, 4), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5)} be a relation on set A. Find equivalence class [1], [2] & [4]

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SOLVED EXAMPLES

Ex.1 Three relation R_1, R_2 and R_3 are defined on set A = {a, b, c} as follows :

(i) R_1 = {a, a), (a, b), (a, c), (b, b), (b, c), (c, a), (c, b), (c, c)}

(ii) R_2 = {(a, b), (b, a), (a, c), (c, a)}

(iii) R_3 = {(a, b), (b, c), (c, a)}

Find whether each of R_1, R_2 and R_3 is reflexive, symmetric and transitive.

Sol. (i) Reflexive : Clearly, (a, a), (b, b), (c, c) $\in R_1$. So, R_1 is reflexive on A.

Symmetric : We observe that (a, b) $\in R_1$ but (b, a) $\notin R_1$. So, R_1 is not symmetric on A.

Transitive : We find that (b, c) $\in R_1$ and (c, a) $\in R_1$ but (b, a) $\notin R_1$. So, R is not transitive on A.

(ii) Reflexive : Since (a, a), (b, b) and (c, c) are not in R_2. So, it is not a reflexive relation on A.

Symmetric : We find that the ordered pairs obtained by interchanging the components of ordered pairs in R_2 are also in R_2. So, R_2 is a symmetric relation on A.

Transitive : Clearly (c, a) $\in R_2$ and (a, b) $\in R_2$ but (c, b) $\notin R_2$. So, it is not a transitive relation on R_2.

(iii) Reflexive : Since non of (a, a), (b, b) and (c, c) is an element of R_3. So, R_3 is not reflexive on A.

Symmetric : Clearly, (b, c) $\in R_3$ but (c, b) $\notin R_3$. so, is not symmetric on A.

Transitive : Clearly, (b, c) $\in R_3$ and (c, a) $\in R_3$ but (b, a) $\notin R_3$. So, R_3 is not transitive on A.

Ex.2 Prove that the relation R on the set Z of all integers defined by (x, y) $\in$ R $\Leftrightarrow$ x – y is divisible by n is an equivalence relation on Z.

Sol. We observe the following properties

Reflexivity : For any a $\in$ N, we have

a – a = 0 = 0 × n $\Rightarrow$ a – a is divisible by n $\Rightarrow$ (a, a) $\in$ R

Thus, (a, a) $\in$ R for all a $\in$ Z

So, R is reflexive on Z

symmetry : Let (a, b) $\in$ R. Then,

(a, b) $\in$ R $\Rightarrow$ (a – b) is divisible by n

$\Rightarrow$ a – b = np for some p $\in$ Z

$\Rightarrow$ b – a = n(–p)

$\Rightarrow$ b – a is divisible by n [$\because$ p $\in$ Z $\Rightarrow$ –p $\in$ Z]

$\Rightarrow$ (b, a) $\in$ R

Thus, (a , b) $\in$ R $\Rightarrow$ (b, a) $\in$ R for all a, b, $\in$ Z

So, R is symmetric on Z.

Transitivity : Let a, b, c $\in$ Z such that (a, b) $\in$ R and (b, c) $\in$ R. Then,

(a , b) $\in$ R $\Rightarrow$ (a – b) is divisible by n

$\Rightarrow$ a – b = np for some p $\in$ Z

(b, c) $\in$ R $\Rightarrow$ (b – c) is divisible by n

$\Rightarrow$ b – c = nq for some q $\in$ Z

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$\therefore$ (a, b) $\in$ R and (b, c) $\in$ R

$\Rightarrow$ a – b = np and b – c – nq

$\Rightarrow$ (a – b) + (b – c) = np + nq

$\Rightarrow$ a – c = n(p + q)

$\Rightarrow$ a – c is divisible by n [$\because$ p, q $\in$ Z $\Rightarrow$ p + q = Z]

$\Rightarrow$ (a, c) $\in$ R

thus, (a, b) $\in$ R and (b, c) $\in$ R $\Rightarrow$ (a, c) $\in$ R for all a, b, c $\in$ Z. so, R is transitive relation in Z.

Ex.3 Show that the relation 'is congruent to' on the set of all triangles in a plane is an equivalence relation.

Sol. Let S be the set of all triangles in a plane and let R be the relation on S defined by $(\Delta_1, \Delta_2) \in R \Leftrightarrow$ triangle Δ_1 is congruent to triangle Δ_2. We observe the following properties.

Reflexivity : For each triangle $\Delta \in S$, we have

$\Delta \cong \Delta \Rightarrow (\Delta, \Delta) \in R$ for all $\Delta \in S \Rightarrow$ R is reflexive on S

Symmetry : Let $\Delta_1, \Delta_2 \in S$ such that $(\Delta_1, \Delta_2) \in R$.

Then, $(\Delta_1, \Delta_2) \in R \Rightarrow \Delta_1 \cong \Delta_2 \Rightarrow \Delta_2 \cong \Delta_1 \Rightarrow (\Delta_2, \Delta_1) \in R$

So, R is symmetric on S

Transitivity : Let $\Delta_1, \Delta_2, \Delta_3 \in S$ such that $(\Delta_1, \Delta_2) \in R$ and $(\Delta_2, \Delta_3) \in R$. Then,

$(\Delta_1, \Delta_2) \in R$ and $(\Delta_2, \Delta_3) \in R \Rightarrow \Delta_1 \cong \Delta_2$ and $\Delta_2 \cong \Delta_3 \Rightarrow \Delta_1 \cong \Delta_3 \Rightarrow (\Delta_1, \Delta_3) \in R$

So, R is transitive on S.

Hence, R being reflexive, symmetric and transitive, is an equivalence relation on S.

Ex.4 (a,a) $\in$ R, for every a $\in$ A. This condition is for which of the following relations ?

(A) Reflexive relation (B) Symmetric relation

(C) Equivalence relation (D) Transitive relation

Ans. (A)

Sol. The above is the condition for a reflexive relation. A relation is said to be reflexive if every element in the set is related to itself.

Ex.5 $(a_1,a_2) \in R$ implies that $(a_2,a_1) \in R$, for all $a_1,a_2 \in A$. This condition is for which of the following relations ?

(A) equivalence relation (B) Reflexive relation

(C) Symmetric relation (D) Universal relation

Ans. (C)

Sol. The above is a condition for a symmetric relation

For example, a relation R on set A = {1,2,3,4} is given by R = {(a,b) : a + b = 3, a > 0 , b > 0}

1 + 2 = 3, 1 > 0 and 2 > 0 which implies (1,2) $\in$ R.

Similarly, 2 + 1 = 3, 2 > 0 and 1 > 0 which implies (2,1) $\in$ R. Therefore both (1,2) and (2,1) are converse of each other and is a part of the relation. Hence, they are symmetric.

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Ex.6 Which of the following relations is symmetric and transitive but not reflexive for the set I = {4,5} ?

(A) R = {(4,4), (5,4), (5,5)}

(B) R = {(4,4), (5,5)}

(C) R = {(4,5), (5,4)}

(D) R = {(4,5), (5,4), (4,4)}

Ans. (D)

Sol. R = {(4,5), (5,4), (4,4)} is symmetric since (4,5) and (5,4) are converse of each other thus satisfying the condition for a symmetric relation and it is transitive as (4,5) $\in$ R and (5,4) $\in$ R implies that (4,4) $\in$ R. It is not reflexive as every element in the set I is not related to itself.

Ex.7 Which of the following relations is transitive but not reflexive for the set S = {3, 4, 6} ?

(A) R = {(3, 4), (4, 6), (3, 6)}

(B) R = {(1, 2), (1, 3), (1, 4)}

(C) R = {(3, 3), (4, 4), (6, 6)}

(D) R = {(3, 4), (4, 3)}

Ans. (A)

Sol. For the above given set S = {3, 4, 6}, R = {(3, 4), (4, 6), (3, 6)} is transitive as (3, 4) $\in$ R and (4, 6) $\in$ R and (3, 6) also belongs to R. It is not a reflexive relation as it does not satisfy the condition (a, a) $\in$ R, for every a $\in$ A for a relation R in the set A.

Ex.8 Which of the following relations is symmetric but neither reflexive nor transitive for a set A = {1, 2, 3}.

(A) R = {(1, 2), (1, 3), (1, 4)}

(B) R = {(1, 2), (2, 1)}

(C) R = {(1, 1), (2, 2), (3, 3)}

(D) R = {(1, 1), (1, 2), (2, 3)}

Ans. (B)

Sol. A relation in a set A is said to be symmetric if $(a_1, a_2) \in R$ implies that $(a_1, a_2) \in R$, for every $a_1, a_2 \in R$.

Hence, for the given set A = {1, 2, 3}, R = {(1, 2), (2, 1)} is symmetric. It is not reflexive since every element is not related to itself and neither transitive as it does not satisfy the condition that for a given relation R in a set A if $(a_1, a_2) \in R$ and $(a_2, a_3) \in R$ implies that $(a_1, a_3) \in R$ for every $a_1, a_2, a_3 \in R$

Ex.9 Find the identity relation on the set P = {x, y, z}.

Sol. We know that the relation (I) is identity, then all the elements of set P are related with itself, such that for every a $\in$ P, then (a, a) $\in$ I.

Given set is P = {x, y, z}

Then the identity relation is given by I = {(x, x), (y, y), (z, z)}.

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Ex.10 Let L be a set of all lines in a XY plane and R be a relation in I defined as R = {(L_1, L_2) : L_1 is parallel to L_2}. what is the type of given relation ?

(A) Reflexive relation (B) Transitive relation

(C) Symmetric relation (D) Equivalence relation

Ans. (D)

Sol. This is an equivalence relation. A relation R is said to be an equivalence relation when it is reflexive, transitive and symmetric.

Reflexive : We know that a line is always parallel to itself. This implies that L_1 is parallel to L_1 i.e. (L_1, L_2) $\in$ R. Hence, it is a reflexive relation.

Symmetric : Now if a line $L_1 || L_2$ then the line $L_2 || L_1$. Therefore, (L_1, L_2) $\in$ R implies that (L_2, L_1) $\in$ R. Hence, it is a symmetric relation.

Transitive : If two lines (L_1, L_3) are parallel to a third line (L_2) then they will be parallel to each other i.e. if (L_1, L_2) $\in$ R and (L_2, L_3) $\in$ R implies that (L_1, L_3) $\in$ R.

Ex.11 Which of the following relations is reflexive but not transitive for the set T = {7, 8, 9} ?

(A) R = {(7, 7), (8, 8), (9, 9), (8,9), (9,7)} (B) R = {(7, 8), (8, 7), (8, 9)}

(C) R = {0} (D) R = {(7, 8), (8, 8), (8, 9)}

Ans. (A)

Sol. The relation R = {(7, 7), (8, 8), (9, 9)} is reflexive as every element is related to itself i.e. (a, a) $\in$ R, for every a $\in$ A and it is not transitive as it does not satisfy the condition that for a relation R in a set A if (a_1, a_2) $\in$ R and (a_2, a_3) $\in$ R implies that (a_1, a_3) $\in$ R for every a_1, a_2, a_3 $\in$ R.

Ex.12 Let S be a set of n > 0 elements. Let B_r be the number of binary relations on S and let B_f be the number of functions from S to S. The expression for B_r and B_f, in terms of n should be

(A) n^2 and $2(n + 1)^2$ (B) n^3 and $n^{(n+1)}$ (C) n and $n^{(n+6)}$ (D) $2^{(n*n)}$ and n^n

Ans. (D)

Sol. For a set with n elements the number of binary relations should be $2^{(n*n)}$ and the number of functions should be n^n.

Hence $B_r = 2^{(n*n)}$ and $B_f = n^n$.

Ex.13 Let P = {1, 2, 3}, R be a relation defined on set P as "is greater than" and R = {(2, 1), (3, 2), (3, 1)}. Verify R is transitive.

Sol. Given set P = {1, 2, 3}

$\therefore$ (3, 2) $\in$ R and (2, 1) $\in$ R

$\Rightarrow$ (3, 1) $\in$ R

This is the only possible pair

$\because$ the relation is transitive

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STATISTICS

1. Statistics

Statistics is a branch of mathematics that deals with the study of collecting, analyzing, interpreting, presenting, and organizing data in a particular manner. Statistics is defined as the process of collection of data, classifying data, representing the data for easy interpretation, and further analysis of data. Statistics also is referred to as arriving at conclusions from the sample data that is collected using surveys or experiments.

1.1 Mathematical Statistics

Statistics is the process of collecting data, evaluating data, and summarizing it into a mathematical form. Initially, statistics were related to the science of the state where it was used in the collection and analysis of facts and data about a country such as its economy, population, etc. Mathematical statistics applies mathematical techniques like linear algebra, differential equations, mathematical analysis, and theories of probability.

There are two methods of analyzing data in mathematical statistics that are used on a large scale:

* Descriptive Statistics
* Inferential Statistics

1.2 Descriptive Statistics

The descriptive method of statistics is used to describe the data collected and summarize the data and its properties using the measures of central tendencies and the measures of dispersion.

1.3 Inferential Statistics

This method of statistics is used to draw conclusions from the data. Inferential statistics requires statistical tests performed on samples, and it draws conclusions by identifying the differences between the 2 groups. Tests calculate the p-value that is compared with the probability of chance(?) = 0.05. If the p-value is less than ?, then it is concluded that the p-value is statistically significant.

1.4 Data Representation in Statistics

The collection of observations and facts is known a data. These observations and facts can be in the form of numbers, measurements, or statements. There are two different kinds of data i.e. Qualitative data and quantitative data. Qualitative data is when the data is descriptive or categorical and quantitative data is when the data is numerical information. Once we know the data collection methods, we aim at representing the collected data in different forms of graphs such as a bar graph, line graph, pie chart, stem and leaf plots, scatter plot, and so on. Before the analysis of data, the outliers are removed that are due to the invariability in the measurements of data. Let us look at different kinds of data representation in statistics.

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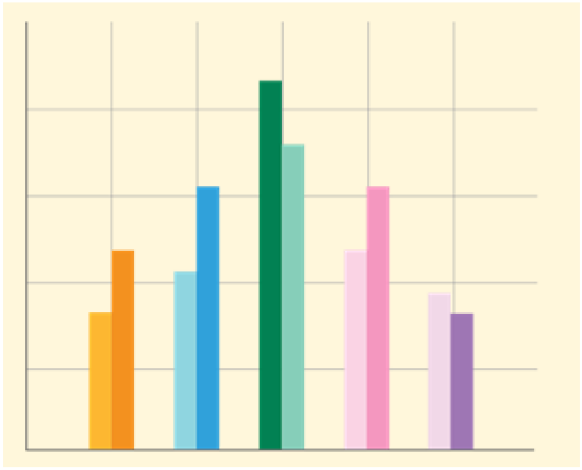
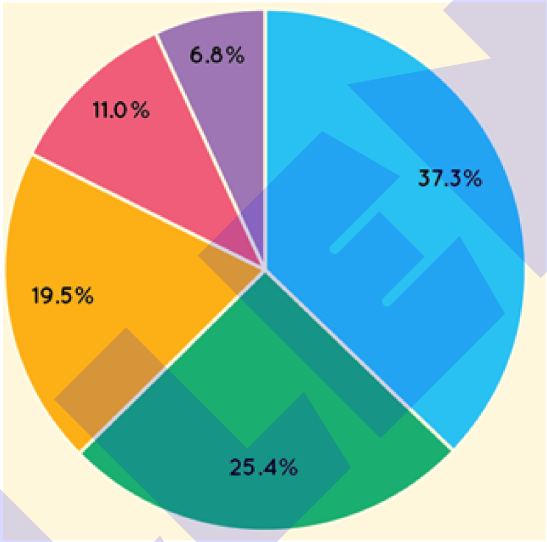


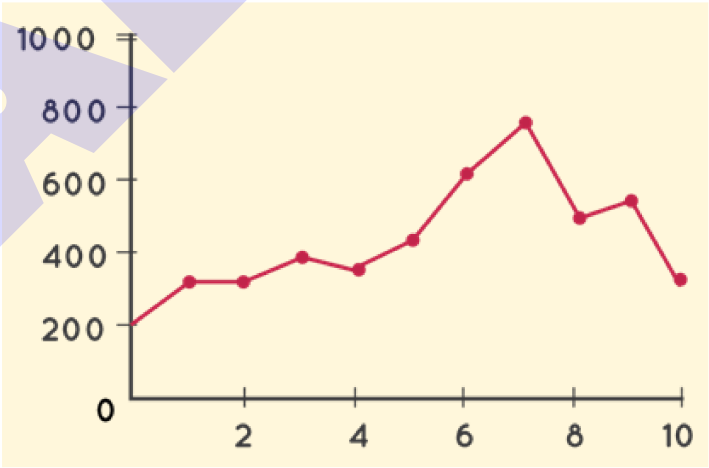
Bar Graph

A group of data represented with rectangular bars with lengths proportional to the values is a bar graph.The bars can either be vertically or horizontally plotted.



Pie Chart

The pie chart is a type of graph in which a circle is divided into Sectors where each sector represents a proportion of the whole.



Line graph

The line graph represents the data in a form of series that is connected with a straight line. These series are called markers.

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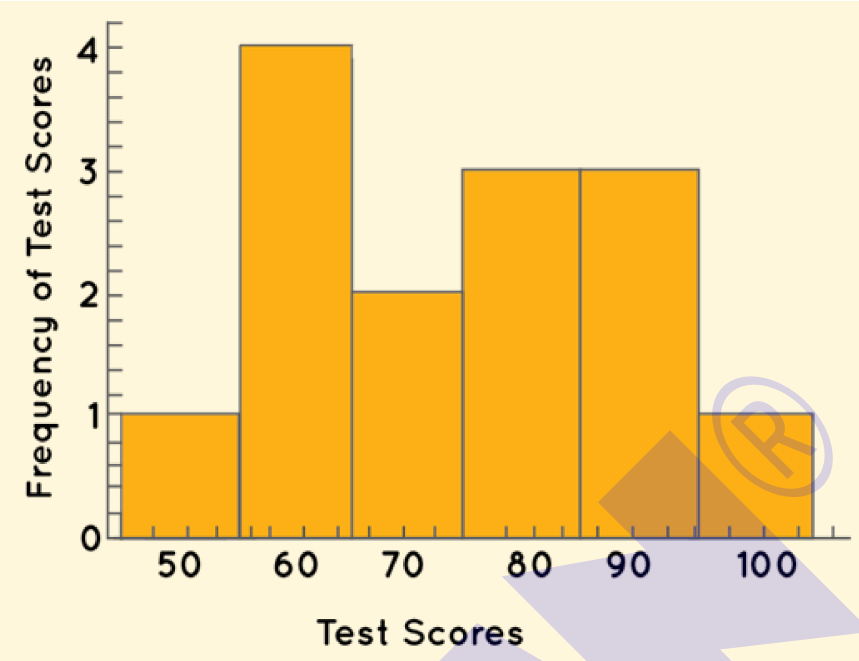

Histogram

The histogram is a type of graph where the diagram consists of rectangles, the area is proportional to the frequency of a variable and the width is equal to the class interval. Here is an example of a histogram.

Height	Frequency
139	1
145	2
150	2
136	1
152	1
144	1
138	2

Frequency Distribution

The frequency distribution table in statistics showcases the data in ascending order along with their corresponding frequencies.The frequency of the data is often represented by f.

Collection of data

The word data means a set of given facts in numerical figures.

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Fundamental characteristics of data

(i) Numerical facts alone form data. Qualitative characteristics, like honesty, poverty, etc., which cannot be measured numerically do not form data.

(ii) Data are aggregate of facts. A single observation does not form data.

(iii)Data collected for a definite purpose may not be suitable for another purpose.

Types of data

(i) **Primary Data :** The data collected by the investigator himself with a definite plan in mind are known as primary data.

(ii) **Secondary Data :** The data collected by someone, other than the investigator, are known as secondary data.

2. Presentation of data

2.1 Raw or ungrouped data

The data obtained in original form are called raw data or ungrouped data.

Ex. The marks obtained by 25 students in a class in a certain examination are given below : 25, 8, 37, 16, 45, 40, 29, 12, 42, 40, 25, 14, 16, 16, 20, 10, 36, 33, 24, 25, 35, 11, 30, 45,48.

This is the raw data.

Array : An arrangement of raw data in ascending or descending order of magnitude is called an array. Arranging the marks of 25 students in ascending order, we get the following array.

8, 10, 11, 12, 14, 16, 16, 16, 20, 24, 25, 25, 25, 29, 30, 33, 35, 36, 37, 40, 40, 42, 45, 45, 48.

2.2 Types of grouped frequency distribution

2.2.1. Exclusive form (or continuous interval form) : A frequency distribution in which the upper limit of each class is excluded and lower limit is included, is called an exclusive form.

Ex. Suppose the marks obtained by some students in an examination are given.

We may consider the classes 0 – 10, 10 – 20 etc. In class 0 – 10, we include 0 and exclude 10. In class 10 - 20, we include 10 and exclude 20.

2.2.2Inclusive form (or discontinuous interval form) : A frequency distribution in which each upper limit as well as lower limit is included, is called an inclusive form. Thus, we have classes of the form 0 – 10, 11 – 20, 21 – 30 etc. In 0 – 10, both 0 and 10 are included.

3. Graphical representation of statistical data

The tabular representation of data is an ideal way of presenting them in a systematic manner. When these numerical figures are represented pictorially or graphically, they become more noticeable and easily intelligible, leaving a more lasting effect on the mind of the observer. With the help of these pictures or graphs, data can be compared easily.

There are various types of graphs. In this chapter, we shall be dealing with the following graphs:

1. Bar graphs 2. Histogram 3. Frequency polygon

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3.1 Bar graph (or column graph or bar chart)

A bar graph is a pictorial representation of numerical data in the form of rectangles (or bars) of equal width and varying heights.

These rectangles are drawn either vertically or horizontally.

The height of a bar represents the frequency of the corresponding observation. The gap between two bars is kept the same.

3.2 Histogram

A histogram is a graphical representation of a frequency distribution in an exclusive form in the form of rectangles with class intervals as bases and the corresponding frequencies as heights, there being no gap between any two successive rectangles.

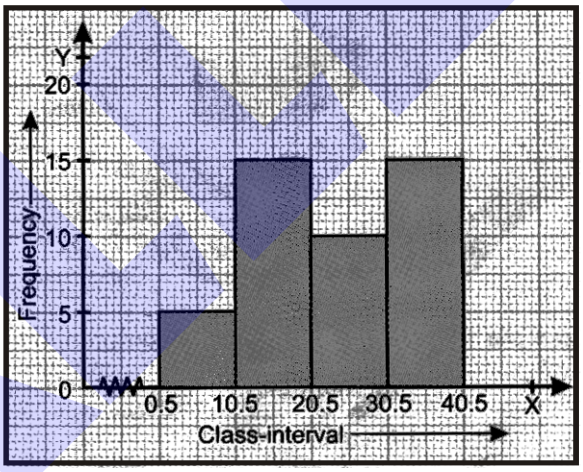
Method of drawing a histogram

Step 1 : If the given frequency distribution is in inclusive form, convert it into an exclusive form.

Step 2 : Taking suitable scales, mark the class-intervals along x-axis and frequencies along y-axis. Note that the scales chosen for both the axes need not be the same.

Step 3 : Construct rectangles with class-intervals as bases and the corresponding frequencies as heights.

Note : Since the scale on x-axis starts at 0.5, a kink is shown near the origin.



A random survey of the number of children of various age groups playing in a park was found as follows.

Age (in years)	1–2	2–3	3–5	5–7	7–10	10–15	15–17
Number of Children	5	3	6	12	9	10	4

Draw a historgram to represent the data above.

Explanation

In the given frequency distribution, we see that the class size are different. Hence, we calculate the adjusted frequency for each class by using the formula.

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Adjusted frequency for a class = $\frac{\text{Minimum class size}}{\text{Class size of the class}} \times$ frequency of the class

In the given distribution, the minimum class size is 1.

The adjusted frequencies are computed in the following table.

Age (in years)	Frequency	Class size	Adjusted frequency
1–2	5	1	$\frac{1}{1} \times 5 = 5$
2–3	3	1	$\frac{1}{1} \times 3 = 3$
3–5	6	2	$\frac{1}{2} \times 6 = 3$
5–7	12	2	$\frac{1}{2} \times 12 = 6$
7–10	9	3	$\frac{1}{3} \times 9 = 3$
10–15	10	5	$\frac{1}{5} \times 10 = 2$
15–17	4	2	$\frac{1}{2} \times 4 = 2$

3.3 Frequency polygon

Let $x_1, x_2, ..., x_n$ be the class marks (i.e., mid points) of the given frequency distribution and let $f_1, f_2, ... f_n$ be the corresponding frequencies. We plot the points $(x_1, f_1), (x_2, f_2),..., (x_n, f_n)$ on a graph paper and join these points by line segments. We complete the diagram in the form of a polygon by taking two more classes (called imagined classes), one at the beginning and the other at the end, each with frequency zero. This polygon is known as the frequency polygon of the given frequency distribution.

Illustration 1 : The following table shows the number of diabetic patients at various age groups.

Age (in years)	10-20	20-30	30-40	40-50	50-60	60-70
Number of diabetic patients	3	6	14	8	5	2

Represent the above data by a frequency polygon.

Solution : Take two more intervals one at the beginning and the other at the end, each with frequency 0. Thus, we have class-intervals

0 – 10, 10 – 20, 20 – 30, 30 – 40, 40 – 50, 50 – 60, 60 – 70 and 70 – 80 with corresponding frequencies as 0, 3, 6, 14, 8, 5, 2, 0 respectively. Thus, we have :

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Class mark	5	15	25	35	45	55	65	75
Frequency	0	3	6	14	8	5	2	0

Now, plot the points (5, 0), (15, 3), (25, 6), (35, 14), (45, 8), (55, 5), (65, 2), (75, 0) on a graph paper and join them successively to get the required graph.

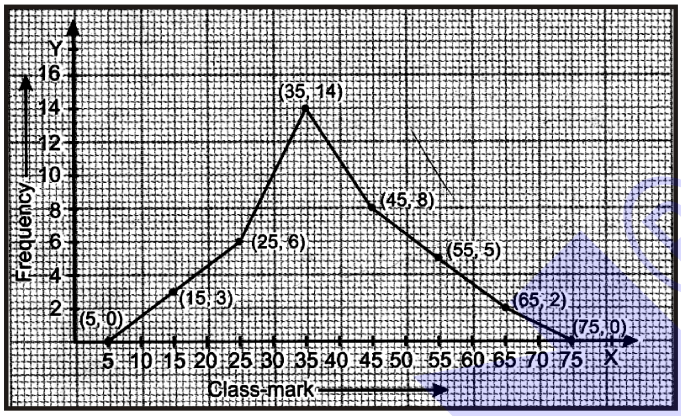


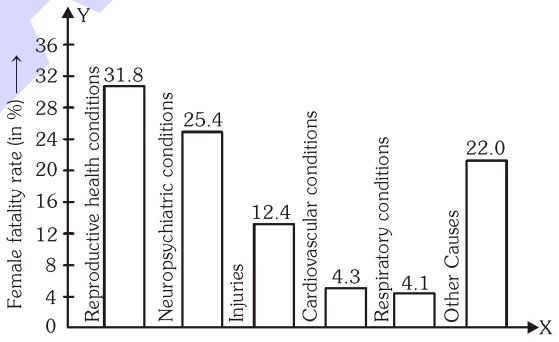
Illustration 2 : A survey conducted by an organisation for the cause of illness and death among the women between the ages 15 – 44 (in years) worldwide, found the following figures (in %):

S.No.	Causes	Female fatality rate (%)
1	Reproductive health conditions	31.8
2	Neuropsychiatric conditions	25.4
3	Injuries	12.4
4	Cardiovascular conditions	4.3
5	Respiratory conditions	4.1
6	Other causes	22

(i) Represent the information given above graphically.

(ii) Which condition is the major cause of women's ill health and death worldwide ?

Solution : (i)



(ii) Reproductive health conditions is the major cause of women's ill health and death worldwide.

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Do yourself - 1 :

1. Given below are the seats won by different political parties in the polling outcome of a state assembly elections:

Political Party	A	B	C	D	E	F
Seats Won	75	55	37	29	10	37

(i) Draw a bar graph to represent the polling results.

(ii) Which political party won the maximum number of seats?

2. The length of 40 leaves of a plant are measured correct to one millimetre, and the obtained data is represented in the following table:

Length (in mm)	Number of leaves
118-126	3
127-135	5
136-144	9
145-153	12
154-162	5
163-171	4
172-180	2

(i) Draw a histogram to represent the given data.

(ii) Is there any other suitable graphical representation for the same data?

(iii) Is it correct to conclude that the maximum number of leaves are 153 mm long? Why?

4. MEASURES OF CENTRAL TENDENCY :

An average value or a central value of a distribution is the value of variable which is representative of the entire distribution, this representative value is called the measures of central tendency. Generally the following five measures of central tendency.

(a) Mathematical average

(i) Arithmetic mean (ii) Geometric mean (iii) Harmonic mean

(b) Positional average

(i) Median (ii) Mode

4.1 ARITHMETIC MEAN :

(i) For ungrouped dist. : If $x_1, x_2, \ldots\ldots x_n$ are n values of variate x_i then their A.M. $\bar{x}$ is defined as

$$\bar{x} = \frac{x_1 + x_2 + \ldots\ldots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\Rightarrow \quad \sum_{i=1}^{n} x_i = n\bar{x}$$

(ii) For ungrouped and grouped freq. dist. : If $x_1, x_2, \ldots. x_n$ are values of variate with corresponding frequencies $f_1, f_2, \ldots f_n$ then their A.M. is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \ldots. + f_nx_n}{f_1 + f_2 + \ldots. + f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N}, \quad \text{where } N = \sum_{i=1}^{n} f_i$$

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(iii) By short method : If the value of x_i are large, then the calculation of A.M. by using previous formula is quite tedious and time consuming. In such case we take deviation of variate from an arbitrary point a.

Let $d_i = x_i - a$

$\therefore \ \bar{x} = a + \dfrac{\Sigma f_i d_i}{N}$, where a is assumed mean

(iv) By step deviation method : Sometime during the application of short method of finding the A.M. If each deviation d_i are divisible by a common number h(let)

Let $u_i = \dfrac{d_i}{h} = \dfrac{x_i - a}{h}$

$\therefore \quad \bar{x} = a + \left(\dfrac{\Sigma f_i u_i}{N}\right) h$

(v) Weighted mean : If $w_1, w_2, \ldots\ldots w_n$ are the weights assigned to the values $x_1, x_2, \ldots\ldots x_n$ respectively then their weighted mean is defined as

$$\text{Weighted mean} = \frac{w_1x_1 + w_2x_2 + \ldots\ldots + w_nx_n}{w_1 + \ldots\ldots + w_n} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

(vi) Combined mean : If $\bar{x}_1$ and $\bar{x}_2$ be the means of two groups having n_1 and n_2 terms respectively then the mean (combined mean) of their composite group is given by

$$\text{combined mean} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

If there are more than two groups then, $\text{combined mean} = \dfrac{n_1\bar{x}_1 + n_1\bar{x}_2 + n_3\bar{x}_3 + \ldots.}{n_1 + n_2 + n_3 + \ldots.}$

(vii) Properties of Arithmetic mean :

- Sum of deviations of variate from their A.M. is always zero i.e. $\Sigma(x_i - \bar{x}) = 0, \ \Sigma f_i(x_i - \bar{x}) = 0$
- Sum of square of deviations of variate from their A.M. is minimum i.e. $\Sigma(x_i - \bar{x})^2$ is minimum
- If $\bar{x}$ is the mean of variate x_i then
  A.M. of $(x_i + \lambda) = \bar{x} + \lambda$
  A.M. of $(\lambda x_i) = \lambda\bar{x}$
  A.M. of $(ax_i + b) = a\bar{x} + b$ (where λ, a, b are constant)
- A.M. is independent of change of assumed mean i.e. it is not effected by any change in assumed mean.

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Illustration 3 : Find the A.M. of the following frequency distribution.

x_i	5	8	11	14	17
f_i	4	5	6	10	20

Solution : Here $N = \Sigma f_i = 4 + 5 + 6 + 10 + 20 = 45$

$\Sigma f_i x_i = (5 \times 4) + (8 \times 5) + (11 \times 6) + (14 \times 10) + (17 \times 20) = 606$

$$\therefore \ \bar{x} = \frac{\Sigma f_i x_i}{N} = \frac{606}{45} = 13.47$$

Illustration 4 : Find the mean of the following frequency distribution

x_i	5	15	25	35	45	55
f_i	12	18	27	20	17	6

Solution : Let assumed mean a = 35, h = 10

here $N = \Sigma f_i = 100$, $u_i = \frac{(x_i - 35)}{10}$

$\therefore \ \Sigma f_i u_i = [12 \times (-3) + 18 \times (-2) + 27 \times (-1) + 20 \times 0 + 17 \times 1 + 6 \times 2] = -70$

$$\therefore \ \bar{x} = a + \left(\frac{\Sigma f_i u_i}{N}\right) h = 35 + \frac{(-70)}{100} \times 10 = 28$$

Illustration 5 : Find the weighted mean of first n natural numbers when their weights are equal to their squares respectively

Solution : Weighted Mean

$$= \frac{1.1^2 + 2.2^2 + ... + n.n^2}{1^2 + 2^2 + ... + n^2} = \frac{1^3 + 2^3 + ... + n^3}{1^2 + 2^2 + ... + n^2} = \frac{[n(n+1)/2]^2}{[n(n+1)(2n+1)/6]} = \frac{3n(n+1)}{2(2n+1)}$$

Illustration 6 : The mean income of a group of persons is Rs. 400 and another group of persons is Rs. 480. If the mean income of all the persons of these two groups is Rs. 430 then find the ratio of the number of persons in the groups.

Solution : Here $\bar{x}_1 = 400$, $\bar{x}_2 = 480$, $\bar{x} = 430$

$$\because \ \bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \Rightarrow 430 = \frac{400 n_1 + 480 n_2}{n_1 + n_2}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{5}{3}$$

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Illustration 7 : Find the mean marks from the following data :

Marks	No. of Students
Below 10	5
Below 20	9
Below 30	18
Below 40	29
Below 50	45
Below 60	60
Below 70	70
Below 80	78
Below 90	83
Below 100	85

Solution : We may prepare the table as given below :

Marks	**No. of students**	**Class Interval**	f_i	**Class mark (x_i)**	f_ix_i
Below 10	5	0-10	5	5	25
Below 20	9	10-20	4	15	60
Below 30	18	20-30	9	25	225
Below 40	29	30-40	11	35	385
Below 50	45	40-50	16	45	720
Below 60	60	50-60	15	55	825
Below 70	70	60-70	10	65	650
Below 80	78	70-80	8	75	600
Below 90	83	80-90	5	85	425
Below 100	85	90-100	2	95	190
			N = 85		Σf_ix_i = 4105

$$\therefore \text{ Mean }, \bar{x} = \frac{\sum f_i x_i}{N} = \frac{4105}{85} = 48.29$$

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Do yourself - 2 :

1. If in an examination different weights are assigned to different subjects Physics (2), Chemistry (1), English (1), Mathematics (2) A student scores 60 in Physics, 70 in Chemistry, 70 in English and 80 in Mathematics, then weighted mean is-

(A) 60 (B) 70 (C) 80 (D) 85

2. If a variable takes the value 0, 1, 2.....n with frequencies proportional to the bionomial coefficients ${}^nC_0, {}^nC_1,.....,{}^nC_n$ then the mean of the distribution is-

(A) $\frac{n(n+1)}{4}$ (B) $\frac{n}{2}$ (C) $\frac{n(n-1)}{2}$ (D) $\frac{n(n+1)}{2}$

3. A student obtained 75%, 80%, 85% marks in three subjects. If the marks of another subject are added then his average marks can not be less than-

(A) 60% (B) 65% (C) 80% (D) 90%

4. If the mean of the series $x_1, x_2, ..., x_n$ is $\bar{x}$, then the mean of the series $x_i + 2i$, i = 1, 2, ..., n will be-

(A) $\bar{x}+n$ (B) $\bar{x}+n+1$ (C) $\bar{x}+2$ (D) $\bar{x}+2n$

5. The mean of the following freq. table is 50 and $\Sigma f = 120$

class	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
f	17	f_1	32	f_2	19

the missing frequencies are-

(A) 28, 24 (B) 24, 36 (C) 36, 28 (D) None of these

4.2 Graphical representation of cumulative frequency distribution

- **Cumulative frequency polygon curve (Ogive)**

Cumulative frequency is of two types and corresponding to these, the ogive is also of two types.

- **Less than series** - **More than series**

- **Less than series :** To construct a cumulative frequency polygon and an ogive, we follow these steps :

Step-1 : Mark the upper class limit along x-axis and the corresponding cumulative frequencies along y-axis.

Step-2 : Plot these points successively by line segments. We get a polygon, called cumulative frequency polygon.

Step-3 : Plot these points successively by smooth curves, we get a curve called cumulative frequency curve or an ogive.

- **More than series :** To construct a cumulative frequency polygon and an ogive, we follow these steps:

Step-1 : Mark the lower class limits along x-axis and the corresponding cumulative frequencies along y-axis.

Step-2 : Plot these points successively by line segments, we get a polygon, called cumulative frequency polygon.

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Step-3 : Plot these points successively by smooth curves, we get a curve, called cumulative frequency curve or an ogive.

Application of an ogive

Ogive can be used to find the median of a frequency distribution. To find the median, we follow these steps.

Method-I

Step-1 : Draw anyone of the two types of frequency curves on the graph paper.

Step-2 : Compute $\frac{N}{2}(N = \Sigma f_i)$ and mark the corresponding points on the y-axis.

Step-3 : Draw a line parallel to x-axis from the point marked in step 2, cutting the cumulative frequency curve at a point P.

Step-4 : Draw perpendicular PM from P on the x-axis. The x - coordinate of point M gives the median.

Method-II

Step-1 : Draw less than type and more than type cumulative frequency curves on the graph paper.

Step-2 : Mark the point of intersecting (P) of the two curves drawn in step 1.

Step-3 : Draw perpendicular PM from P on the x-axis. The x- coordinate of point M gives the median.

Illustration 8 : The following distribution gives the daily income of 50 workers of a factory.

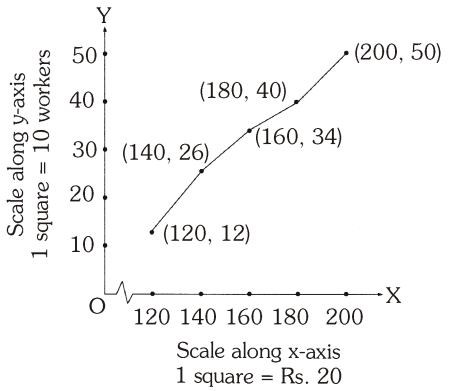
Daily income (in Rs.)	100-120	120-140	140-160	160-180	180-200
No. of workers	12	14	8	6	10

Convert the distribution above to a less than type cumulative frequency distribution and draw its ogive.

Solution : From the given table, we prepare a less than type cumulative frequency distribution table, as given below:

Income less than (in Rs.)	120	140	160	180	200
Cumulative frequency	12	26	34	40	50

Now, plot the points (120, 12), (140, 26), (160, 34), (180, 40) and (200, 50).
Join these points by a freehand curve to get an ogive of 'less than' type.



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Do yourself - 3 :

1. The following table gives the weight of 120 articles :

Weight (in kg)	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Number of students	14	17	22	26	23	18

Change the distribution to a 'more than type' distribution and draw its ogive.

2. The annual profits earned by 30 shops of a shopping complex in a locality gives rise to the following distribution:

Profit (in lakhs Rs.)	No. of shops (frequency)
More than or equal to 5	30
More than or equal to 10	28
More than or equal to 15	16
More than or equal to 20	14
More than or equal to 25	10
More than or equal to 30	7
More than or equal to 35	3

Draw both ogives for the data above. Hence, obtain the median profit.

4.3 MEDIAN :

The median of a series is the value of middle term of the series when the values are written in ascending order. Therefore median, divided an arranged series into two equal parts.

Formulae of median :

(i) For ungrouped distribution : Let n be the number of variate in a series then

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{term}, \text{ (when n is odd)} \\ \text{Mean of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+1\right)^{th} \text{ terms, (when n is even)} \end{cases}$$

(ii) For ungrouped freq. dist. : First we prepare the cumulative frequency (c.f.) column and Find value of N then

$$\text{Median} = \begin{cases} \left(\frac{N+1}{2}\right)^{th} \text{term}, \text{ (when N is odd)} \\ \text{Mean of } \left(\frac{N}{2}\right)^{th} \text{ and } \left(\frac{N}{2}+1\right)^{th} \text{ terms, (when N is even)} \end{cases}$$

(iii) For grouped freq. dist : Prepare c.f. column and find value of $\frac{N}{2}$ then find the class which contain value of c.f. is equal or just greater to N/2, this is median class

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$$\therefore \text{ Median} = \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h$$

where ℓ — lower limit of median class
f — freq. of median class
F — c.f. of the class preceeding median class
h — Class interval of median class

Illustration 9 : Find the median of following freq. dist.

class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
f	8	30	40	12	10

Solution :

class	f_i	c.f.
0 – 10	8	8
10 – 20	30	38
20 – 30	40	78
30 – 40	12	90
40 – 50	10	100

Here $\frac{N}{2} = \frac{100}{2} = 50$ which lies in the value 78 of c.f. hence corresponding class of this c.f. is 20-30 is the median class, so $\ell = 20$, f = 40, F = 38, h = 10

$$\therefore \text{ Median} = \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h = 20 + \frac{(50-38)}{40} \times 10 = 23$$

4.4 MODE :

In a frequency distribution the mode is the value of that variate which have the maximum frequency

Method for determining mode :

(i) For ungrouped dist. : The value of that variate which is repeated maximum number of times

(ii) For ungrouped freq. dist. : The value of that variate which have maximum frequency.

(iii) For grouped freq. dist. : First we find the class which have maximum frequency, this is model calss

$$\therefore \text{ Mode} = \ell + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h$$

where ℓ — lower limit of model class
f_0 — freq. of the model class
f_1 — freq. of the class preceeding model class
f_2 — freq. of the class succeeding model class
h — class interval of model class

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Illustration 10 : Find the mode of the following frequency distribution

class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
f_i	2	18	30	45	35	20	6	3

Solution : Here the class 30–40 has maximum freq. so this is the model class

$\ell = 30, f_0 = 45, f_1 = 30, f_2 = 35, h = 10$

$$\therefore \text{Mode} = \ell + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h = 30 + \frac{45 - 30}{2 \times 45 - 30 - 35} \times 10 = 36$$

Do yourself - 4 :

1. Given below is the frequency distribution of the heights of players in a school.

Heights (in cm)	160-162	163-165	166-168	169-171	172-174
No. of students	15	118	142	127	18

Find the average height of maximum number of students.

2. The mode of the following series is 36. Find the missing frequency f in it.

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	8	10	f	16	12	6	7

4.5 RELATION BETWEEN MEAN, MEDIAN AND MODE :

In a moderately asymmetric distribution following relation between mean, median and mode of a distribution. It is known as imprical formula.

Mode = 3 Median – 2 Mean

Note (i) Median always lies between mean and mode

(ii) For a symmetric distribution the mean, median and mode are coincide.

5. MEASURES OF DISPERSION :

The dispersion of a statistical distribution is the measure of deviation of its values about the their average (central) value.

It gives an idea of scatteredness of different values from the average value.

Generally the following measures of dispersion are commonly used.

(i) Range (ii) Mean deviation (iii) Variance and standard deviation

5.1 RANGE : The difference between the greatest and least values of variate of a distribution, are called the range of that distribution.

If the distribution is grouped distribution, then its range is the difference between upper limit of the maximum class and lower limit of the minimum class.

$$\text{Also, coefficient of range} = \frac{\text{difference of extreme values}}{\text{sum of extreme values}}$$

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5.2 Mean deviation (M.D.) : The mean deviation of a distribution is, the mean of absolute value of deviations of variate from their statistical average (Mean, Median, Mode).

If A is any statistical average of a distribution then mean deviation about A is defined as

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} |x_i - A|}{n} \qquad \text{(for ungrouped dist.)}$$

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} f_i |x_i - A|}{N} \qquad \text{(for freq. dist.)}$$

Note :- Mean deviation is minimum when it taken about the median

$$\text{Coefficient of Mean deviation} = \frac{\text{Mean deviation}}{A}$$

(where A is the central tendency about which Mean deviation is taken)

Illustration 11 : Find the range of following numbers 10, 8, 12, 11, 14, 9, 6

Solution : Greatest value = 14, Least value = 6

Range = 14 – 6 = 8

Illustration 12 : Find the mean deviation of number 3, 4, 5, 6, 7

Solution : Here n = 5, $\bar{x} = 5$

$$\therefore \quad \text{Mean deviation} = \frac{\Sigma |x_i - \bar{x}|}{n}$$

$$= \frac{1}{5}[|3-5| + |4-5| + |5-5| + |6-5| + |7-5|]$$

$$= \frac{1}{5}[2 + 1 + 0 + 1 + 2] = \frac{6}{5} = 1.2$$

Illustration 13 : Find the mean deviation about mean from the following data

x_i	3	9	17	23	27
f_i	8	10	12	9	5

Solution :

x_i	f_i	$f_i x_i$	$\lvert x_i - \bar{x} \rvert$	$f_i \lvert x_i - \bar{x} \rvert$
3	8	24	12	96
9	10	90	6	60
17	12	204	2	24
23	9	207	8	72
27	5	135	12	60
	$N = 44$	$\Sigma f_i x_i = 660$		$\Sigma f_i \lvert x_i - \bar{x} \rvert = 312$

$$\text{Mean } (\bar{x}) = \frac{\Sigma f_i x_i}{N} = \frac{660}{44} = 15$$

$$\text{Mean deviation} = \frac{\Sigma f_i |x_i - \bar{x}|}{N} = \frac{312}{44} = 7.09$$

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Limitations of mean deviation : In a series, where the degree of variability is very high, the median is not a representative central tendency. Thus, the mean deviation about median calculated for which series can not be fully relied.

The sum of the deviations from the mean (minus signs igonored) is more than the sum of the deviations from median. Therefore, the mean deviation about the mean is not very scientific. Thus, in many cases, mean deviation may give unsatisfactory results. Also mean deviation is calculated on the basis of absolute values of the deviations and therefore, cannot be subjected to further algebraic treatment. This implies that we must have some other measure of dispersion. Standard deviation is such a measure of dispersion.

5.3 Variance and standard deviation : The variance of a distribution is, the mean of squares of deviation of variate from their mean. It is denoted by σ^2 or var(x).

The positive square root of the variance are called the standard deviation. It is denoted by σ or S.D.

Hence standard deviation $= +\sqrt{\text{variance}}$

Formulae for variance :

(i) for ungrouped dist. :

$$\sigma_x^2 = \frac{\Sigma(x_i - \bar{x})^2}{n}$$

$$\sigma_x^2 = \frac{\Sigma x_i^2}{n} - \bar{x}^2 = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2$$

$$\sigma_d^2 = \frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2, \quad \text{where } d_i = x_i - a$$

(ii) For freq. dist. :

$$\sigma_x^2 = \frac{\Sigma f_i(x_i - \bar{x})^2}{N}$$

$$\sigma_x^2 = \frac{\Sigma f_i x_i^2}{N} - (\bar{x})^2 = \frac{\Sigma f_i x_i^2}{N} - \left(\frac{\Sigma f_i x_i}{N}\right)^2$$

$$\sigma_d^2 = \frac{\Sigma f_i d_i^2}{N} - \left(\frac{\Sigma f_i d_i}{N}\right)^2$$

$$\sigma_u^2 = h^2\left[\frac{\Sigma f_i u_i^2}{N} - \left(\frac{\Sigma f_i u_i}{N}\right)^2\right] \quad \text{where } u_i = \frac{d_i}{h}$$

Note :- $\sigma^2 = \sigma_x^2 = \sigma_d^2 = h^2\sigma_u^2$

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5.4. MATHEMATICAL PROPERTIES OF VARIANCE :

- $\text{Var.}(x_i + \lambda) = \text{Var.}(x_i)$

  $\text{Var.}(\lambda x_i) = \lambda^2.\text{Var}(x_i)$

  $\text{Var}(ax_i + b) = a^2.\text{Var}(x_i)$

  where λ, a, b, are constant

Illustration 14 : Find the variance of first n natural numbers

Solution : $\sigma^2 = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2 = \frac{\Sigma n^2}{n} - \left(\frac{\Sigma n}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n} - \left\{\frac{n(n+1)}{2n}\right\}^2 = \frac{n^2-1}{12}$

Illustration 15 : If $\sum_{i=1}^{18}(x_i - 8) = 9$ and $\sum_{i=1}^{18}(x_i - 8)^2 = 45$, then find the standard deviation of $x_1, x_2, \ldots. x_{18}$

Solution : Let $(x_i - 8) = d_i$

$$\therefore \sigma_x = \sigma_d = \sqrt{\frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2} = \sqrt{\frac{45}{18} - \left(\frac{9}{18}\right)^2} = \sqrt{\frac{5}{2} - \frac{1}{4}} = \frac{3}{2}$$

Illustration 16 : Variance of the data given below is

Size of item	3.5	4.5	5.5	6.5	7.5	8.5	9.5
Frequency	3	7	22	60	85	32	8

(1) 1.29 (2) 2.19 (3) 1.32 (4) none of these

Solution : Let the assumed mean be a = 6.5

Calculation of variance

x_i	f_i	$d_i = x_i - 6.5$	$f_i d_i$	$f_i d_i^2$
3.5	3	–3	–9	27
4.5	7	–2	–14	28
5.5	22	–1	–22	22
6.5	60	0	0	0
7.5	85	1	85	85
8.5	32	2	64	128
9.5	8	3	24	72
	$N = \Sigma f_i = 217$		**$\Sigma f_i d_i = 128$**	**$\Sigma f_i d_i^2 = 362$**

Here N = 217, $\Sigma f_i d_i = 128$ and $\Sigma f_i d_i^2 = 362$

$$\therefore \text{Var}(X) = \left(\frac{1}{N}\Sigma f_i d_i^2\right) - \left(\frac{1}{N}\Sigma f_i d_i\right)^2 = \frac{362}{217} - \left(\frac{128}{217}\right)^2 = 1.668 - 0.347 = 1.321$$

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Do yourself - 5 :

1. The variance of first 20-natural numbers is-

(A) $\frac{133}{4}$ (B) $\frac{379}{12}$ (C) $\frac{133}{2}$ (D) $\frac{399}{4}$

2. The mean deviation about median from the following data 340, 150, 210, 240, 300, 310, 320, is-

(A) 52.4 (B) 52.5 (C) 52.8 (D) none of these

3. The mean and variance of 5 observations of an experiment are 4 and 5.2 respectively. If from these observations three are 1, 2 and 6, then the remaining will be-

(A) 2, 9 (B) 5, 6 (C) 4, 7 (D) 3, 8

4. The mean deviation of the series a, a + d, a + 2d, ..., a + 2nd from its mean is-

(A) $\frac{n+1}{2n+1}|d|$ (B) $\frac{n(n+1)}{2n+1}|d|$ (C) $\frac{n(n-1)}{2n+1}|d|$ (D) none of these

5. Let $x_1, x_2 ..... x_n$ be values taken by a variable X and $y_1, y_2, ..., y_n$ be the values taken by a variable Y such that $y_i = ax_i + b$, $i = 1, 2, ..., n$. Then-

(A) Var (Y) = a^2Var(X) (B) Var(Y) = a^2Var(X) + b

(C) Var(Y) = Var(X) + b (D) None of these

6. The mean square deviation of a set of n observations $x_1, x_2, ..., x_n$ about a point c is defined as $\frac{1}{n}\sum_{i=1}^{n}(x_i - c)^2$. The mean square deviation about –2 and 2 are 18 and 10 respectively, then standard deviation of this set of observations is-

(A) 3 (B) 2 (C) 1 (D) None of these

5.5 MEAN SQUARE DEVIATION :

The mean square deviation of a distrubution is the mean of the square of deviations of variate from assumed mean. It is denoted by S^2

Hence $S^2 = \frac{\Sigma(x_i - a)^2}{n} = \frac{\Sigma d_i^2}{n}$ (for ungrouped dist.)

$S^2 = \frac{\Sigma f_i(x_i - a)^2}{N} = \frac{\Sigma f_i d_i^2}{N}$ (for freq. dist.), where $d_i = (x_i - a)$

5.6 RELATION BETWEEN VARIANCE AND MEAN SQUARE DEVIATION :

$\because \quad \sigma^2 = \frac{\Sigma f_i d_i^2}{N} - \left(\frac{\Sigma f_i d_i}{N}\right)^2$

$\Rightarrow \sigma^2 = s^2 - d^2$, where $d = \bar{x} - a = \frac{\Sigma f_i d_i}{N}$

$\Rightarrow s^2 = \sigma^2 + d^2 \Rightarrow s^2 \geq \sigma^2$

Hence the variance is the minimum value of mean square deviation of a distribution

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Illustration 17 : Determine the variance of the following frequency dist.

class	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12
f_i	2	7	12	19	9	1

Solution : Let a = 7, h = 2

class	x_i	f_i	$u_i = \frac{x_i - a}{h}$	$f_i u_i$	$f_i u_i^2$
0 – 2	1	2	–3	–6	18
2 – 4	3	7	–2	–14	28
4 – 6	5	12	–1	–12	12
6 – 8	7	19	0	0	0
8 – 10	9	9	1	9	9
10 – 12	11	1	2	2	4
		$N = 50$		$\Sigma f_i u_i = -21$	$\Sigma f_i u_i^2 = 71$

$$\therefore \sigma^2 = h^2\left[\frac{\Sigma f_i u_i^2}{N} - \left(\frac{\Sigma f_i u_i}{N}\right)^2\right] = 4\left[\frac{71}{50} - \left(\frac{-21}{50}\right)^2\right] = 4[1.42 - 0.1764] = 4.97$$

Note : Combined variance of two data

- If means of two series containing n_1, n_2 terms are $\bar{x}_1, \bar{x}_2$ and their variance's are σ_1^2, σ_2^2 respectively and their combined mean is $\bar{x}$ then the variance σ^2 of their combined series is given by following formula

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)} \quad \text{where } d_i = \bar{x}_1 - \bar{x},\ d_2 = \bar{x}_2 - \bar{x}$$

i.e. $$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$$

Illustration 18 : The mean of two groups of sizes 200 and 300 are 25 and 10 respectively.Their standard deviation are 3 and 4 respectively. The variance of combined sample of size 500 is-

(1) 64 (2) 65.2 (3) 67.2 (4) 64.2

Solution : Combined mean $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{200 \times 25 + 300 \times 10}{500} = 16$

Here $d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$ and $d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$

We know that

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} = \frac{200(9 + 81) + 300(16 + 36)}{500} = \frac{33600}{500} = 67.2$$

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Illustration 19 : The mean and variance of a series containing 5 terms are 8 and 24 respectively. The mean and variance of another series containing 3 terms are also 8 and 24 respectively. The variance of their combined series will be-

(1) 20 (2) 24 (3) 25 (4) 42

Solution : Using $\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\overline{x}_1 - \overline{x}_2)^2$

$$\Rightarrow \sigma^2 = \frac{5(24) + 3(24)}{5+3} + \frac{5(3)}{(5+3)^2}(8-8)^2 = 24$$

5.7 Analysis of Frequency Distributions

In earlier sections, we have studied about some types of measures of dispersion. The mean deviation and the standard deviation have the same units in which the data are given. Whenever we want to compare the variability of two series with same mean, which are measured in different units, we do not merely calculate the measures of dispersion but we require such measures which are independent of the units. The measure of variability which is independent of units is called coefficient of variation (denoted as C.V.)

The coefficient of variation is defined as

$$\text{C.V.} = \frac{\sigma}{x} \times 100, \overline{x} \neq 0,$$

where σ and $\overline{x}$ are the standard deviation and mean of the data.

For comparing the variability or dispersion of two series, we calcualte the coefficient of variation for each series. The series having greater C.V. is said to be more variable than the other. The series having lesser C.V. is said to be more consistent than the other.

Comparison of two frequency distributions with same mean Let $\overline{x}_1$ and σ_1 be the mean and standard deviation of the first distribution, and $\overline{x}_2$ and σ_2 be the mean and standard deviation of the second distribution.

Then $\text{C.V. (1st distribution)} = \frac{\sigma_1}{\overline{x}_1} \times 100$

and $\text{C.V. (2nd distribution)} = \frac{\sigma_2}{\overline{x}_2} \times 100$

Given $\overline{x}_1 = \overline{x}_2 = \overline{x}$ (say)

Therefore $\text{C.V. (2st distribution)} = \frac{\sigma_1}{\overline{x}} \times 100$ ...(1)

and $\text{C.V. (2nd distribution)} = \frac{\sigma_2}{\overline{x}} \times 100$ ...(2)

It is clear from (1) and (2) that the two C.Vs. can be compared on the basis of values of σ_1 and σ_2 only. Thus, we say that for two series with equal means, the series with greater standard deviation (or variance) is called more variable or dispersed than the other Also the series with lesser value of standard deviation (or variance) is said to be more consistent than the other

Note : Coefficient of S.D. $= \frac{\sigma}{\overline{x}}$

Coefficient of variation $= \frac{\sigma}{\overline{x}} \times 100$ (in percentage)

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Illustration 20 : Find the coefficient of variation of first n natural numbers

Solution : For first n natural numbers.

$$\text{Mean } (\bar{x}) = \frac{n+1}{2}, \text{ S.D.}(\sigma) = \sqrt{\frac{n^2-1}{12}}$$

$$\therefore \text{ coefficient of variance} = \frac{\sigma}{\bar{x}} \times 100 = \sqrt{\frac{n^2-1}{12}} \times \frac{1}{\left(\frac{n+1}{2}\right)} \times 100 = \sqrt{\frac{(n-1)}{3(n+1)}} \times 100$$

Illustration 21 : Two plants A and B of a factory show following results about the number of workers and the wages paid to them.

	A	B
No. of workers	5000	6000
Average monthly wages	Rs 2500	Rs 2500
Variance of distribution of wages	81	100

In which plant, A or B is there greater variability in individual wages ?

Solution : The variance of the distribution of wages in plant A$(\sigma_1^2) = 81$

Therefore, standard deviation of the distribution of wages in plant A $(\sigma_1) = 9$

Also, the variance of the distribution of wages in plant B$(\sigma_2^2) = 100$

Therefore, standard deviation of the distribution of wages in plant B$(\sigma_2) = 10$

Since the average monthly wages in both the plants is same, i.e., Rs. 2500, therefore, the plant with greater standard deviation will have more variability.

Thus, the plant B has greater variability in the individual wages.

Illustration 22 : Coefficient of variation of two distributions are 60 and 70, and their standard deviations are 21 and 16, respectively. What are their arithmetic means.

Solution : Given C.V. (1st distribution) = 60, $\sigma_1 = 21$

C.V. (2nd distribution) = 70, $\sigma_2 = 16$

Let $\bar{x}_1$ and $\bar{x}_2$ be the means of 1st and 2nd distribution, respectively. Then

$$\text{C.V. (1st distribution)} = \frac{c_1}{\bar{x}_1} \times 100$$

Therefore $$60 = \frac{21}{\bar{x}_1} \times 100 \text{ or } \bar{x}_1 = \frac{21}{60} \times 100 = 35$$

and $$\text{C.V.}\left(\text{2nd distribution}\right) = \frac{c_2}{\bar{x}_2} \times 100$$

i.e. $$70 = \frac{16}{\bar{x}_2} \times 100 \text{ or } \bar{x}_2 = \frac{16}{70} \times 100 = 22.85$$

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Do yourself - 6 :

1. The following values are calculated in respect of heights and weights of the students of a section of Class XI :

	heights	weights
Mean	162.6 cm	52.36 kg
Variance	127.69 cm^2	23.1361 kg^2

can we say that the weight show greater variation than the heights ?

2. From the data given below state which group is more variable, A or B ?

Marks	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Group A	9	17	32	33	40	10	9
Group B	10	20	30	25	43	15	7

3. From the prices of shares X and Y below, find out which is more stable in value :

X	35	54	52	53	56	58	52	50	51	49
Y	108	107	105	105	106	107	104	103	104	101

6. MATHEMATICAL EXPECTATION :

If 'p' represents a person's chance of success in any venture and 'M' the sum of money which he will receive in case of success, then the sum of money denoted by pM is called his expectation.

Example : Let X represent the outcome of a roll of a fair die. The possible values for X are 1, 2, 3, 4, 5 and 6, all of which are equally likely with a probability of $\frac{1}{6}$. The expectation of X is

$$E[X] = 1.\frac{1}{6} + 2.\frac{1}{6} + 3.\frac{1}{6} + 4.\frac{1}{6} + 5.\frac{1}{6} + 6.\frac{1}{6} = 3.5$$

7. PROBABILITY DISTRIBUTION :

Let S be a sample space. A random variable X is a function from the set S to R, the set of real numbers. A Probability Distribution spells out how a total probability of 1 is distributed over several values of a random variable.

For example, the sample space for a throw of a pair of dice is

$$S = \begin{matrix} \{(1,1), & (1,2), & \dots, & (1,6) \\ (2,1), & (2,2), & \dots, & (2,6) \\ \vdots & & & \\ (6,1), & (6,2), & \dots, & (6,6)\} \end{matrix}$$

Let X be the sum of numbers on the dice. Then X(1, 2) = 3, X(4, 3) = 7, etc. Also, {X = 7} is the event {(6, 1), 5, 2), (4, 3), (3, 4), (2, 5), (1, 6)}. In general, if X is a random variable defined on the sample space S and r is a real number, then {X = r} is an event.

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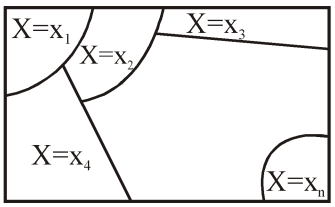
If the random variable X takes n distinct values $x_1, x_2, ..., x_n$, then $\{X = x_1\}, \{X = x_2\}, ..., \{X = x_n\}$ are mutually exclusive and exhaustive events.



Now, since $(X = x_i)$ is an event, we can talk of $P(X = x_i)$. If $P(X = x_i) = P_i (1 \le i \le n)$, then the system of numbers $\begin{pmatrix} x_1 & x_2 & ... & x_n \\ p_1 & p_2 & ... & p_n \end{pmatrix}$ is said to be the probability distribution of the random variable X.

Mean of any probability distribution of a random variable is given by :

$$\mu = \frac{\sum p_i x_i}{\sum p_i} = \sum p_i x_i \quad (\text{Since } \Sigma p_i = 1)$$

Variance of a random variable is given by, $\sigma^2 = \sum (x_i - \mu)^2 . p_i$

$$\sigma^2 = \sum p_i x_i^2 - \mu^2$$

(Note that Standard Deviation (SD) = $+\sqrt{\sigma^2}$)

7.1 Binomial probability distribution :

The probability distribution for a binomial variate 'X' is given by ; $P(X= r)= {}^nC_r\, p^r\, q^{n-r}$

where : p = probability of success in a single trial, q = probability of failure in a single trial and p + q = 1.

The recurrence formula $\frac{P(r+1)}{P(r)} = \frac{n-r}{r+1} \cdot \frac{p}{q}$, is very helpful for quickly computing P(1), P(2) . P(3) etc. if P(0) is known.

Mean and variance of the binomial distribution :

Let the binomial probability distribution is

X	0	1	2	...	n
P(X)	${}^nC_0 q^n p^0$	${}^nC_1 q^{n-1} p$	${}^nC_2 q^{n-2} p^2$	...	${}^nC_n q^0 p^n$

The mean of this distribution is $\sum_{i=1}^{n} X_i p_i = \sum_{X=1}^{n} X . {}^nC_X q^{n-X} p^X = np$,

The variance of the Binomial distribution is $\sigma^2 = npq$ and the standard deviation is $\sigma = \sqrt{(npq)}$.

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8. POISSON'S PROBABILITY DISTRIBUTION :

In BPD when $n \to \infty$ and $p \to 0$ such that $np = \lambda$, is a finite quantity then BPD can be approximated to PPD.

In BPD, $P(X = r) = {}^nC_r p^r q^{n-r}$

$$\lim_{\substack{n\to\infty \\ p\to 0}} \frac{n(n-1)(n-2).....(n-r+1)}{r!} p^r (1-p)^{n-r} = \lim_{\substack{n\to\infty \\ p\to 0}} \frac{n(n-1)(n-2).....(n-r+1).\lambda^r}{n^r.r!} \frac{(1-(\lambda/n))^n}{(1-p)^r}$$

$$= \lim_{n\to\infty} \frac{1\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right).....\left(1-\frac{r-1}{n}\right)}{r!}.\lambda^r.\frac{\left[(1-(\lambda/n))^{-n/\lambda}\right]^{-\lambda}}{\left(1-\frac{\lambda}{n}\right)^r} = \frac{\lambda^r}{r!}.e^{-\lambda}$$

Hence $P(x = r) = \frac{e^{-\lambda}\lambda^r}{r!}$, where $np = \lambda$ and λ is called the parameter of P.P.D.

Note :

(i) Mean of PPD = 1
(ii) Variance of PPD = 1
(iii) SD of PPD = $\sqrt{\lambda}$

$\lambda = np$

Illustration 23 : A random variable X has the probability distribution:

X :	1	2	3	4	5	6	7	8
p(X) :	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events E = {X is a prime number} and F = {X < 4}, the probability $P(E \cup F)$ is -

(1) 0.35 (2) 0.77 (3) 0.87 (4) 0.50

Solution : E = x is a prime number

$P(E) = P(2) + P(3) + P(5) + P(7) = 0.62$

$F = (x < 4)$, $P(F) = P(1) + P(2) + P(3) = 0.50$

$\therefore$ $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$= 0.62 + 0.50 - 0.35 = 0.77$

Illustration 24 : The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is -

(1) $\frac{128}{256}$ (2) $\frac{219}{256}$ (3) $\frac{37}{256}$ (4) $\frac{28}{256}$

Solution : $\left.\begin{matrix} np = 4 \\ npq = 2 \end{matrix}\right\} \Rightarrow q = \frac{1}{2}, p = \frac{1}{2}, n = 8$

$$P(X = 2) = {}^8C_2\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^6 = 28.\frac{1}{2^8} = \frac{28}{256}$$

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Do yourself - 7 :

1. A die is tossed twice. Getting a number greater than 4 is considered a success. Then the variance of the probability distribution of the number of successes is

(A) $\frac{2}{9}$ (B) $\frac{4}{9}$ (C) $\frac{1}{3}$ (D) None of these

2. A die is tossed thrice. A success is getting 1 or 6 on a toss. The mean and the variance of number of successes

(A) $\mu = 1, \sigma^2 = \frac{2}{3}$ (B) $\mu = \frac{2}{3}, \sigma^2 = 1$

(C) $\mu = 2, \sigma^2 = \frac{2}{3}$ (D) None of these

3. The mean and variance of a binomial distribution are 6 and 4. The parameter n is

(A) 18 (B) 12 (C) 10 (D) 9

4. The mean and variance of a binomial distribution are 4 and 3 respectively, then the probability of getting exactly six successes in this distribution is

(A) ${}^{16}C_6\left(\frac{1}{4}\right)^{10}\left(\frac{3}{4}\right)^{6}$ (B) ${}^{16}C_6\left(\frac{1}{4}\right)^{6}\left(\frac{3}{4}\right)^{10}$

(C) ${}^{12}C_6\left(\frac{1}{4}\right)^{10}\left(\frac{3}{4}\right)^{6}$ (D) ${}^{12}C_6\left(\frac{1}{4}\right)^{6}\left(\frac{3}{4}\right)^{6}$

5. The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is

(A) $\frac{28}{256}$ (B) $\frac{219}{256}$ (C) $\frac{128}{256}$ (D) $\frac{37}{256}$

6. Two cards are drawn successively with replacement from a well shuffled deck of 52 cards then the mean of the number of aces is

(A) $\frac{1}{13}$ (B) $\frac{3}{13}$ (C) $\frac{2}{13}$ (D) None of these

7. In a binomial distribution, the mean is 4 and variance is 3. Then its mode is

(A) 6 (B) 5 (C) 4 (D) None of these

8. A dice is thrown 100 times and getting an even number is considered a success. The variance of the number of successes will be

(A) 10 (B) 25 (C) 18 (D) 10

9. A random variable X has the following probability ditribution

$X = x_i$	1	2	3	4
$P(X = x_i)$	0.1	0.2	0.3	0.4

The mean and the standard deviation are respectively

(A) 3 and 2 (B) 3 and 1 (C) 3 and $\sqrt{3}$ (D) 2 and 1

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MATHEMATICAL REASONING

1. STATEMENT :

A sentence which is either true or false but cannot be both are called a statement. A sentence which is an exclamatory or a wish or an imperative or an interrogative can not be a statement.

If a statement is true then its truth value is T and if it is false then its truth value is F

For ex.

(i) "New Delhi is the capital of India", a true statement

(ii) "3 + 2 = 6", a false statement

(iii) "Where are you going ?" not a statement beasuse

it connot be defined as true or false

Note : A statement cannot be both true and false at a time

2. SIMPLE STATEMENT :

Any statement whose truth value does not depend on other statement are called simple statement

For ex. (i) "$\sqrt{2}$ is an irrational number" (ii) "The set of real number is an infinite set"

3. COMPOUND STATEMENT :

A statement which is a combination of two or more simple statements are called compound statement

Here the simple statements which form a compound statement are known as its sub statements

For ex.

(i) "If x is divisible by 2 then x is even number"

(ii) "ΔABC is equilatral if and only if its three sides are equal"

4. LOGICAL CONNECTIVES :

The words or phrases which combined simple statements to form a compound statement are called logical connectives.

In the following table some possible connectives, their symbols and the nature of the compound statement formed by them

S.N.	Connectives	symbol	use	operation
1.	and	$\wedge$	$p \wedge q$	conjunction
2.	or	$\vee$	$p \vee q$	disjunction
3.	not	$\sim$ or '	$\sim p$ or p'	negation
4.	If .... then	$\Rightarrow$ or $\rightarrow$	$p \Rightarrow q$ or $p \rightarrow q$	Implication or conditional
5.	If and only if (iff)	$\Leftrightarrow$ or $\leftrightarrow$	$p \Leftrightarrow q$ or $p \leftrightarrow q$	Equivalence or Bi-conditional

Explanation :

(i) $p \wedge q \equiv$ statement p and q

($p \wedge q$ is true only when p and q both are true otherwise it is false)

(ii) $p \vee q \equiv$ statement p or q

($p \vee q$ is true if at least one from p and q is true i.e. $p \vee q$ is false only when p and q both are false)

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(iii) $\sim p \equiv$ not statement p

($\sim p$ is true when p is false and $\sim p$ is false when p is true)

(iv) $p \Rightarrow q \equiv$ statement p then statement q

($p \Rightarrow q$ is false only when p is true and q is false otherwise it is true for all other cases)

(v) $p \Leftrightarrow q \equiv$ statement p if and only if statement q

($p \Leftrightarrow q$ is true only when p and q both are true or false otherwise it is false)

5. TRUTH TABLE :

A table which shows the relationship between the truth value of compound statement S(p, q, r ....) and the truth values of its sub statements p, q, r, ... is said to be truth table of compound statement S

If p and q are two simple statements then truth table for basic logical connectives are given below

Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Negation

p	$(\sim p)$
T	F
F	T

Conditional

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$ or $p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Note : If the compound statement contain n sub statements then its truth table will contain 2^n rows.

6. LOGICAL EQUIVALENCE :

Two compound statements $S_1(p, q, r...)$ and $S_2(p, q, r ....)$ are said to be logically equivalent or simply equivalent if they have same truth values for all logically possibilities

Two statements S_1 and S_2 are equivalent if they have identical truth table i.e. the entries in the last column of their truth table are same. If statements S_1 and S_2 are equivalent then we write $S_1 \equiv S_2$

For ex. The truth table for $(p \rightarrow q)$ and $(\sim p \vee q)$ given as below

p	q	$(\sim p)$	$p \rightarrow q$	$\sim p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

We observe that last two columns of the above truth table are identical hence compound statements $(p \rightarrow q)$ and $(\sim p \vee q)$ are equivalent

i.e. $\boxed{p \rightarrow q \equiv \sim p \vee q}$

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7. TAUTOLOGY AND CONTRADICTION :

(i) Tautology : A statement is said to be a tautology if it is true for all logical possibilities i.e. its truth value always T. it is denoted by t.

For ex. the statement $p \vee \sim (p \wedge q)$ is a tautology

p	q	$p \wedge q$	$\sim (p \wedge q)$	$p \vee \sim (p \wedge q)$
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

Clearly, The truth value of $p \vee \sim (p \wedge q)$ is T for all values of p and q. so $p \wedge \sim (p \wedge q)$ is a tautology

(ii) Contradiction : A statement is a contradiction if it is false for all logical possibilities.

i.e. its truth value always F. It is denoted by c.

For ex. The statement $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction

p	q	$\sim p$	$\sim q$	$p \vee q$	$(\sim p \wedge \sim q)$	$(p \vee q) \wedge (\sim p \wedge \sim q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

Clearly, then truth value of $(p \vee q) \wedge (\sim p \wedge \sim q)$ is F for all value of p and q. So $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction.

Note : The negation of a tautology is a contradiction and negation of a contradiction is a tautology

8. DUALITY :

Two compound statements S_1 and S_2 are said to be duals of each other if one can be obtained from the other by replacing $\wedge$ by $\vee$ and $\vee$ by $\wedge$

If a compound statement contains the special variable t (tautology) and c (contradiction) then obtain its dual we replaced t by c and c by t in addition to replacing $\wedge$ by $\vee$ and $\vee$ by $\wedge$.

Note :

(i) the connectives $\wedge$ and $\vee$ are also called dual of each other.

(ii) If $S^*(p, q)$ is the dual of the compound statement $S(p, q)$ then

(a) $S^*(\sim p, \sim q) \equiv \sim S(p, q)$ (ii) $\sim S^*(p, q) \equiv S(\sim p, \sim q)$

For ex. The duals of the following statements

(i) $(p \wedge q) \vee (r \vee s)$ (ii) $(p \vee t) \wedge (p \vee c)$

(iii) $\sim(p \wedge q) \vee [p \wedge \sim(q \vee \sim s)]$

are as given below

(i) $(p \vee q) \wedge (r \wedge s)$

(ii) $(p \wedge c) \vee (p \wedge t)$

(iii) $\sim(p \vee q) \wedge [p \vee \sim(q \wedge \sim s)]$

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9. CONVERSE, INVERSE AND CONTRAPOSITIVE OF THE CONDITIONAL STATEMENT ($p \to q$):

(i) Converse : The converse of the conditional statement $p \to q$ is defined as $q \to p$

(ii) Inverse : The inverse of the conditional statement $p \to q$ is defined as $\sim p \to \sim q$

(iii) Contrapositive : The contrapositive of conditional statement $p \to q$ is defined as $\sim q \to \sim p$

10. NEGATION OF COMPOUND STATEMENTS :

If p and q are two statements then

(i) **Negation of conjunction :** $\sim(p \wedge q) \equiv \sim p \vee \sim q$

p	q	~ p	~ q	$(p \wedge q)$	~ $(p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

(ii) Negation of disjunction : $\sim(p \vee q) \equiv \sim p \wedge \sim q$

p	q	~ p	~ q	$(p \vee q)$	$(\sim p \vee q)$	$(\sim p \wedge \sim q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

(iii) Negation of conditional : $\sim(p \to q) \equiv p \wedge \sim q$

p	q	~ q	$(p \to q)$	~ $(p \to q)$	$(p \wedge \sim q)$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

(iv) Negation of biconditional : $\sim (p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

we know that $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

$\therefore \quad \sim(p \leftrightarrow q) \equiv \sim[(p \to q) \wedge (q \to p)]$

$\equiv \sim (p \to q) \vee \sim (q \to p)$

$\equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

Note : The above result also can be proved by preparing truth table for $\sim(p \leftrightarrow q)$ and $(p \wedge \sim q) \vee (q \wedge \sim p)$

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11. ALGEBRA OF STATEMENTS :

If p, q, r are any three statements then the some low of algebra of statements are as follow

(i) Idempotent Laws :

(a) $p \wedge p \equiv p$ (b) $p \vee p \equiv p$

i.e. $p \wedge p \equiv p \equiv p \vee p$

p	$(p \wedge p)$	$(p \vee p)$
T	T	T
F	F	F

(ii) Comutative laws :

(a) $p \wedge q \equiv q \wedge p$ (b) $p \vee q \equiv q \vee p$

p	q	$(p \wedge q)$	$(q \wedge p)$	$(p \vee q)$	$(q \vee p)$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	F	F	T	T
F	F	F	F	F	F

(iii) Associative laws :

(a) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

(b) $(p \vee q) \vee r \equiv p \vee (q \vee r)$

p	q	r	$(p \wedge q)$	$(q \wedge r)$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Similarly we can proved result (b)

(iv) Distributive laws : (a) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ (c) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge (p \wedge r)$

(b) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (d) $p \vee (q \vee r) \equiv (p \vee q) \vee (p \vee r)$

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p	q	r	$(q \vee r)$	$(p \wedge q)$	$(p \wedge r)$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Similarly we can prove result (b), (c), (d)

(v) De Morgan Laws : (a) $\sim (p \wedge q) \equiv \sim p \vee \sim q$

(b) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

p	q	$\sim p$	$\sim q$	$(p \wedge q)$	$\sim (p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Similarly we can proved resulty (b)

(vi) Involution laws (or Double negation laws) : $\sim(\sim p) \equiv p$

p	$\sim p$	$\sim (\sim p)$
T	F	T
F	T	F

(vii) Identity Laws : If p is a statement and t and c are tautology and contradiction respectively then

(a) $p \wedge t \equiv p$ (b) $p \vee t \equiv t$ (c) $p \wedge c \equiv c$ (d) $p \vee c \equiv p$

p	t	c	$(p \wedge t)$	$(p \vee t)$	$(p \wedge c)$	$(p \vee c)$
T	T	F	T	T	F	T
F	T	F	F	T	F	F

(viii) Complement Laws :

(a) $p \wedge (\sim p) \equiv c$ (b) $p \vee (\sim p) \equiv t$ (c) $(\sim t) \equiv c$ (d) $(\sim c) \equiv t$

p	$\sim p$	$(p \wedge \sim p)$	$(p \vee \sim p)$
T	F	F	T
F	T	F	T

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(ix) Contrapositive laws : $p \to q \equiv \sim q \to \sim p$

p	q	~ p	~ q	$p \to q$	$\sim q \to \sim p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

12. QUANTIFIED STATEMENTS AND QUANTIFIERS :

The words or phrases "All", "Some", "None", "There exists a" are examples of quantifiers.

A statement containing one or more of these words (or phrases) is a quantified statement.

E.g. (1) All dogs are poodles

(2) Some books have hard covers

(3) There exists an odd number which is prime.

Note : Phrases "There exists a" and "Atleast one" and the word "some" have the same meaning.

NEGATION OF QUANTIFIED STATEMENTS :

(1) '**None**' is the negation of '**at least one**' or '**some**' or '**few**'

Statement : Some dogs are poodles.

Negation : No dogs are poodles.

Similarly negation of '**some**' is '**none**'

(2) The negation of "**some A are B**" or "**There exist A which is B**" is "**No A are (is) B**" or "**There does not exist any A which is B".**

Statement-1 : Some boys in the class are smart

Statement-2 : There exists a boy in the class who is smart

Statement-3 :Alteast one boy in the class is smart

All the three above statements have same meaning as they all indicate "**existence**" of smart boy in the class.

Negation of these statements is

No boy in the class is smart.

or

There does not exist any boy in the class who is smart.

(3) Negation of "**All A are B**" is "**Some A are not B**".

Statement : All boys in the class are smart.

Negation : Some boys in the class are not smart.

or

There exists a boy in the class who is not smart.

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SOLVED EXAMPLES

Ex.1 Which of the following is correct for the statements p and q ?

(1) $p \wedge q$ is true when at least one from p and q is true

(2) $p \to q$ is true when p is true and q is false

(3) $p \leftrightarrow q$ is true only when both p and q are true

(4) $\sim (p \vee q)$ is true only when both p and q are false

Sol.(4) We know that $p \wedge q$ is true only when both p and q are true so option (1) is not correct

we know that $p \to q$ is false only when p is true and q is false so option (2) is not correct

we know that $p \leftrightarrow q$ is true only when either p and q both are true or both are flase

so option (3) is not correct

we know that $\sim(p \vee q)$ is true only when $(p \vee q)$ is false

i.e. p and q both are false

So option (4) is correct

Ex.2 $\sim(p \vee q) \vee (\sim p \wedge q)$ is equivalent to-

(1) p (2) ~p (3) q (4) ~q

Sol.(2) $\because \sim(p \vee q) \vee (\sim p \wedge q) \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q)$ (By Demorgan Law)

$\equiv \sim p \wedge (\sim q \vee q)$ (By distributive laws)

$\equiv \sim p \wedge t$ (By complement laws)

$\equiv \sim p$ (By Identity Laws)

Ex.3 Which of the following is logically equivalent to $(p \wedge q)$?

(1) $p \to \sim q$ (2) $\sim p \vee \sim q$ (3) $\sim(p \to \sim q)$ (4) $\sim(\sim p \wedge \sim q)$

Sol.(3) $\because p \to \sim q \equiv \sim p \vee \sim q \equiv \sim(p \wedge q)$ $(\because p \to q \equiv \sim p \vee q)$

i.e. $\sim(p \to \sim q) \equiv p \wedge q$

$\because \sim p \vee \sim q \equiv \sim(p \wedge q)$

and $\sim(\sim p \wedge \sim q) \equiv p \vee q$

Ex.4 If $p \to (q \vee r)$ is false, then the truth values of p, q, r respectively are-

(1) T, F, F (2) F, F, F (3) F, T, T (4) T, T, F

Sol.(1) We know $p \to (q \vee r)$ is false only when p is true and $(q \vee r)$ is false. but $(q \vee r)$ is false only when q and r both are false

Hence truth values of p, q, r are respectively T, F, F

Ex.5 Statement $(p \wedge \sim q) \wedge (\sim p \vee q)$ is

(1) a tautology (2) a contradiction

(3) neither a tautology not a contradiction (4) None of these

Sol.(2) $\because (p \wedge \sim q) \wedge (\sim p \vee q)$

$\equiv (p \wedge \sim q) \wedge \sim(p \wedge \sim q)$ (By Demargon Laws)

$\equiv c$, where c is contradiction (By complement laws)

Ex.6 Negation of the statement $p \to (q \wedge r)$ is-

(1) $\sim p \to \sim(q \wedge r)$ (2) $\sim p \vee (q \wedge r)$ (3) $(q \wedge r) \to p$ (4) $p \wedge (\sim q \vee \sim r)$

Sol.(4) $\sim(p \to (q \wedge r)) \equiv p \wedge \sim(q \wedge r)$ $(\because \sim(p \to q) \equiv p \wedge \sim q)$

$\equiv p \wedge (\sim q \vee \sim r)$

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Ex.7 If x = 5 and y = –2 then x – 2y = 9. The contrapositive of this statement is-

(1) If x – 2y ≠ 9 then x ≠ 5 or y ≠ –2 (2) If x – 2y ≠ 9 then x ≠ 5 and y ≠ –2

(3) If x – 2y = 9 then x = 5 and y = –2 (4) None of these

Sol.(1) Let p, q, r be the three statements such that

p : x = 5, q : y = –2 and r : x – 2y = 9

Here given statement is $(p \wedge q) \rightarrow r$ and its contrapositive is $\sim r \rightarrow \sim(p \wedge q)$

i.e. $\sim r \rightarrow (\sim p \vee \sim q)$

i.e. if x – 2y ≠ 9 then x ≠ 5 or y ≠ –2

Ex.8 Which of the following is wrong ?

(1) $p \rightarrow q$ is logically equivalent to $\sim p \vee q$

(2) $(p \vee q) \wedge (q \vee r)$ is true if truth values of p, q, r are T, F, T respectively

(3) $\sim(p \wedge (q \vee r)) \equiv (\sim p \vee \sim q) \wedge (\sim p \vee \sim r)$

(4) The truth value of $p \wedge \sim(p \vee q)$ is always T

Sol.(4) We know that $p \rightarrow q \equiv \sim p \vee q$

If $(p \vee q) \wedge (q \vee r)$ is true then

$(p \vee q)$ and $(q \vee r)$ both are true.

i.e. truth values of p, q, r may be T, F, T respectively

$\because$ $\sim(p \wedge (q \vee r)) \equiv \sim((p \wedge q) \vee (p \wedge r) \equiv \sim(p \wedge q) \wedge \sim(p \wedge r) \equiv (\sim p \vee \sim q) \wedge (\sim p \vee \sim r)$

If p is true and q is false then $\sim(p \vee q)$ is false i.e. $p \wedge \sim(p \vee q)$ is false

Ex.9 If S*(p, q, r) is the dual of the compound statement S(p, q, r) and $S(p, q, r) = \sim p \wedge [\sim(q \vee r)]$ then S*(~p, ~q, ~r) is equivalent to-

(1) S(p, q, r) (2) ~S(~p, ~q, ~r) (3) ~S(p, q, r) (4) S*(p, q, r)

Sol.(3) $\because$ $S(p, q, r) = \sim p \wedge [\sim(q \vee r)]$

So $S(\sim p, \sim q, \sim r) \equiv \sim(\sim p) \wedge [\sim(\sim q \vee \sim r)] \equiv p \wedge (q \wedge r)$

$S^*(p, q, r) \equiv \sim p \vee [\sim(q \wedge r)]$

$S^*(\sim p, \sim q, \sim r) \equiv p \vee (q \vee r)$

Clearly $S^*(\sim p, \sim q, \sim r) \equiv \sim S(p, q, r)$

Ex.10 The negation of the statement "If a quadrilateral is a square then it is a rhombus"

(1) If a quadrilateral is not a square then is a rhombus it

(2) If a quadrilateral is a square then it is not a rhombus

(3) a quadrilateral is a square and it is not a rhombus

(4) a quadritateral is not a square and it is a rhombus

Sol.(3) Let p and q be the statements as given below

p : a quadrilateral is a square

q : a quadritateral is a rhombus

the given statement is $p \rightarrow q$

$\because$ $\sim(p \rightarrow q) \equiv p \wedge \sim q$

Therefore the negation of the given statement is a quadrilateral is a square and it is not a rhombus

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